ABHIJEET

PhD Researcher | Reinforcement Learning | Robot Control | Motion Planning

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Research Profile

PhD researcher working at the intersection of reinforcement learning, robotic control, and trajectory optimization. Develops safe learning and feedback-planning methods, evaluates policies on nonlinear systems, and builds Python research software for simulation and experimentation.

Education

Texas A&M University — College Station, TX
PhD, Aerospace Engineering | GPA: 4.0/4.0 — *Aug 2023–present; expected May 2028*

Indian Institute of Technology Kanpur — Kanpur, India
M.Tech., Aerospace Engineering | GPA: 10.0/10.0 — *Jan 2021–Dec 2022*

Indian Institute of Technology Kanpur — Kanpur, India
B.Tech., Aerospace Engineering | GPA: 8.2/10.0 — *Jul 2017–Dec 2020*

Research Experience

Texas A&M University — College Station, TX
Learning-Based Control Research — *Jun 2025–present*

- Developed PPO and TD3 agents for obstacle navigation with a control barrier function/QP safety filter; compared reward, safety, and computational effort with SAC.
- Evaluated policies across 15 independent training runs and unseen obstacle layouts to study generalization in a nonlinear robotic navigation task.
- Explored behavior cloning for a simulated drone by training a multilayer perceptron to imitate expert control policies, with feedback control applied near expert trajectories.

Texas A&M University — College Station, TX
Graduate Research Assistant – trajectory optimization and robot control — *Aug 2023–present*

- Developed Box-iLQR, combining iterative LQR, logarithmic barriers, and continuation to plan trajectories with bounded states/controls and compute time-varying feedback policies.
- Applied the Python solver to MuJoCo robotic models, including a six-input swimming fish; studied closed-loop performance under noise and disturbances in nonlinear control tasks.
- Developed an SQP interpretation of iLQR and DDP, analyzing iLQR descent and differences in convergence behavior; presented the work at ACC 2026.
- Benchmarked perturbation-feedback solvers for finite- and infinite-horizon nonlinear control against nonlinear-programming and receding-horizon methods.

Research Software

Box-iLQR-Python — 2026–present
Project documentation | Python / MuJoCo — *Open source*

- Implemented state/control bounds, finite-difference derivatives, regularized backward passes, feasibility-preserving line search, and barrier continuation; added convergence diagnostics and automated tests.
- Built simulation examples for nonlinear systems and a six-actuator MuJoCo fish to study feedback trajectory planning.

D2C — 2023–present
Project documentation | Python / MuJoCo — *Open source*

- Developed D2C (Data-to-Control), a data-driven control framework that uses simulator-generated trajectories to learn models/controllers for complex nonlinear dynamical systems.

Selected Publications

- **A Sequential Quadratic Programming Perspective on Optimal Control**. American Control Conference (ACC), 2026. SQP analysis of iLQR and DDP descent and convergence.
- **An Efficient iLQR Algorithm for Optimal Control with Bounded States and Inputs**. AAS Astrodynamics Specialist Conference, 2026. Constrained trajectory optimization and feedback.
- **An Optimal Solution to Infinite Horizon Nonholonomic and Discounted Nonlinear Control Problems**. American Control Conference (ACC), 2025. Perturbation-feedback methods for nonlinear control.
- **Convexity in Optimal Control Problems**. IEEE Conference on Decision and Control (CDC), 2024. Convexity and solution properties in optimal control.
- **Refining GLSDC for Orbit Determination**. AAS Astrodynamics Specialist Conference, 2026. Uncertainty and multiple feasible solution classes from noisy short-arc observations.
- **Box-iLQR: Constraint-Aware Feedback via Finite-Barrier Trajectory Optimization**. Manuscript under review, ICRA 2027.

Technical Skills

Programming	Python, C, C++, MATLAB, Git
ML and scientific computing	PyTorch, NumPy, SciPy, pytest
Research methods	Reinforcement learning (PPO, TD3, SAC), behavior cloning, control barrier functions, iLQR, DDP, motion planning, nonlinear optimization
Robotics and simulation	MuJoCo, ROS/ROS 2, CasADi, Simulink

Additional Research Experience

Indian Institute of Science — Bengaluru, India
Aircraft Controls Researcher — *May–Jul 2023*

- Developed gain-scheduled feedback controllers for nonlinear aircraft dynamics across multiple operating regimes.
- Computed equilibrium conditions and local linearizations for multi-surface aircraft models and designed stabilizing controllers around each operating point.
- Implemented and evaluated the complete controller-scheduling pipeline in Simulink, including transitions between operating regimes.

Graduate Teaching Assistant — Texas A&M University
Graduate Mathematics | Dynamics | Flight Mechanics | Spacecraft Control — *August 2023– Present*

- Supported graduate-level courses in linear algebra, functional analysis, and partial differential equations; graded assignments and conducted office hours/problem-solving sessions.
- Teaching Assistant for Aerospace Dynamics, Spacecraft Control, and Flight Mechanics; supported instruction and problem-solving in Newton–Euler equations, linear systems, spacecraft attitude determination and control, and aircraft flight dynamics and control.

Space Robotics Lab, IIT Kanpur — Kanpur, India
Spacecraft robotics and controls research — *2021–2022*

- Built a three-reaction-wheel experimental platform and a mobile docking testbed; integrated Arduino, ROS, and Simulink for actuation and control experiments.

Selected Honors

- Aero Graduate Excellence Fellowship, Texas A&M University (2024, 2025).
- Aero Graduate Travel Award, Texas A&M University (2024, 2026).
- Institute Silver Medal, Indian Institute of Technology Kanpur (2019).